2.3.14

EE25BTECH11011 - Navya

Question:
If a line makes an angle of 30° with the positive direction of the X-axis, 120° with the positive direction of the Y-axis, then find the angle which it makes with the positive direction of the Z-axis

Solution:
Let the direction (unit) vector of the line be

$$d = \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \tag{1}$$

Condition with x-axis:

$$d^T e_x = \cos 30° \tag{2}$$

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \cos 30° \tag{3}$$

$$a = \frac{\sqrt{3}}{2} \tag{4}$$

Condition with y-axis:

$$d^T e_y = \cos 120° \tag{5}$$

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \cos 120° \tag{6}$$

$$b = -\frac{1}{2} \tag{7}$$

Now using normalization:

$$d^T d = 1 \tag{8}$$

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 1 \tag{9}$$

$$a^2 + b^2 + c^2 = 1 \tag{10}$$

$$\left(\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + c^2 = 1 \tag{11}$$

$$\frac{3}{4} + \frac{1}{4} + c^2 = 1 \tag{12}$$

$$c^2 = 0 \quad \Rightarrow \quad c = 0 \tag{13}$$

Hence,

$$d = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix} \tag{14}$$

Angle with z-axis:

$$d^T e_z = \cos\theta \tag{15}$$

$$\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \cos\theta \tag{16}$$

$$0 = \cos\theta \tag{17}$$

$$\theta = 90^\circ \tag{18}$$

Therefore, the line makes angles of 30°, 120°, and 90° with the x-, y-, and z-axes, respectively.

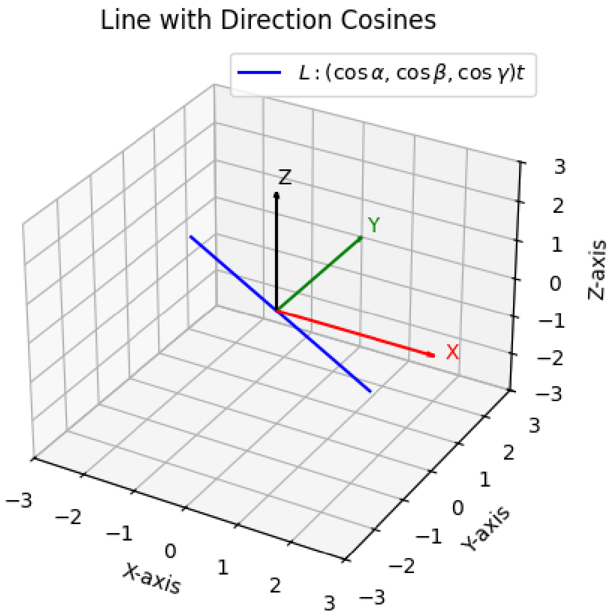

Fig. 0: Direction Cosines